RUNTIME TABLE

Measured computational time and hardware-equivalent scenarios. Eight models across the primary and sensitivity outcomes at the 6- and 8-year horizons. All durations are in minutes.

Model	Actual hardware	Fits	Measured fit	Measured prediction	Measured total	FP32-based estimate	Bandwidth scenario
LR	CPU	524	5.02	0.03	5.06	5.06	5.06
GNB	CPU	404	0.07	0.05	0.12	0.12	0.12
DT	CPU	1,124	0.98	0.06	1.04	1.04	1.04
LightGBM	RTX 5070 Ti	524	4.84	0.09	4.93	4.93	4.93
CatBoost	RTX 3070 Ti	764	13.73	0.07	13.79	6.83	9.36
RealMLP	RTX 3070 Ti	204	67.79	0.07	67.86	33.62	46.05
TabICL	RTX 5070 Ti	204	4.48	62.66	67.14	67.14	67.14
GAT	RTX 5070 Ti / L40S	204	359.37	2.17	361.55	685.56	350.90
Total		3,952	456.28	65.20	521.49	804.29	484.59

Notes. Times sum the four outcome–horizon combinations, including hyperparameter tuning, internal refits, temporal refits and prediction. Fit counts include tuning candidates. These are accumulated task times, not elapsed project time. Imputation, SHAP, permutation importance, bootstrap, queueing and failed or retried work are excluded.

Hardware scenarios. Reference device: RTX 5070 Ti. Equivalent time equals measured total multiplied by source-device capacity divided by reference-device capacity. FP32 and bandwidth columns are separate idealized scenarios; neither is a calibrated same-device measurement or a confidence bound. CPU times are retained unchanged. Host CPU, mixed precision, overhead and actual utilization are not calibrated. L40S has higher theoretical FP32 capacity, so that scenario estimates a longer equivalent duration on RTX 5070 Ti.

Parameters. FP32 capacity (TFLOPS) / bandwidth (GB/s): RTX 3070 Ti, 21.74976 / 608; RTX 5070 Ti, 43.904 / 896; L40S, 91.6 / 864. Specification sources and all 32 individual run summaries are retained in the accompanying workbook.